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COURSE NAME : COMPILER DESIGN FOR CODE ANALYSIS

COURSE CODE : CSA1323

1. DESIGN A LEXICAL ANALYZER TO VALIDATE OPERATORS TO RECOGNIZE THE OPERATORS +,-,*,/ USING REGULAR ARITHMETIC OPERATORS .

C PROGRAMMING CODE:

```
#include <stdio.h>
#include <string.h>

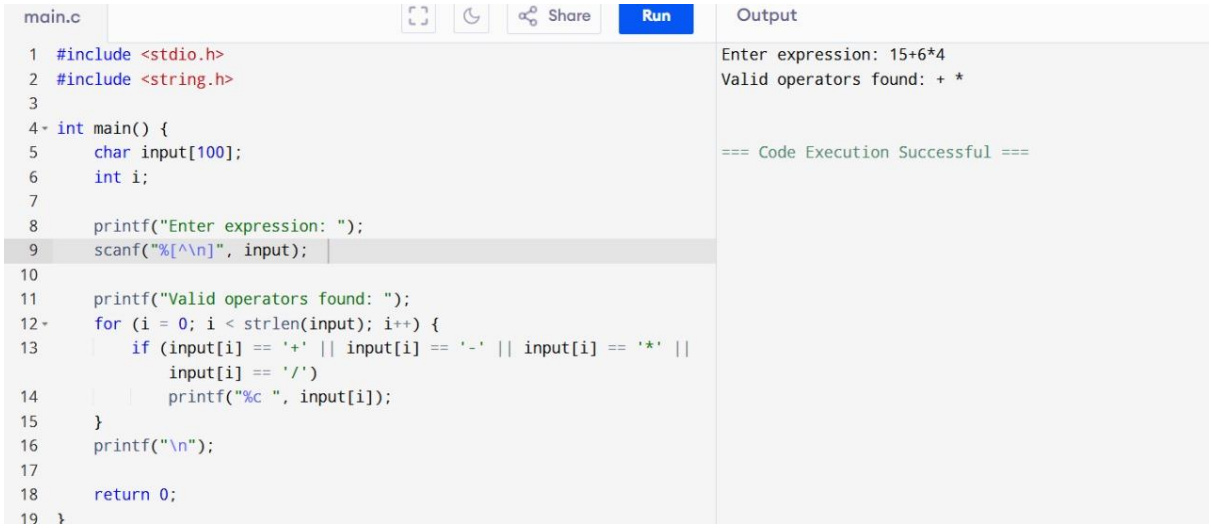
int main() {
    char input[100];
    int i;

    printf("Enter expression: ");
    scanf("%s", input);

    printf("Valid operators found: ");
    for (i = 0; i < strlen(input); i++) {
        if (input[i] == '+' || input[i] == '-' || input[i] == '*' || input[i] == '/')
            printf("%c ", input[i]);
    }
    printf("\n");

    return 0;
}
```

OUTPUT:



2. DESIGN A LEXICAL ANALYZER TO FIND THE NUMBER OF WHITESPACES AND NEWLINE CHARACTERS.

C PROGRAMMING CODE:

```
#include <stdio.h>
int main() {
    char ch;
    int spaces = 0, newlines = 0;

    printf("Enter text (end input with #):\n");
    while ((ch = getchar()) != '#') {
        if (ch == ' ')
            spaces++;
        else if (ch == '\n')
            newlines++;
    }

    printf("\nNumber of whitespaces: %d\n", spaces);
```

```
    printf("Number of newlines: %d\n", newlines);
    return 0;
}
```

OUTPUT:

